

Mosfet Amplifier

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Introduction

The purpose of this project is to design a MOSFET amplifier using the theory from lectures. By working on a single part of the amplifier (stage), then combining them to create a single functional amplifier, with each respective stage having a purpose. The amplifier must also meet the outlined design specifications.

Design

After iterations, I ended with a three-stage amplifier. The first and second stages are common sources with a current source load, while the third stage is a source follower. The goal of the first and second stages is to achieve high gain, whereas the third stage is implemented to reduce load gain losses.

Biasing: Current Mirror

Before we implement the stages, we can design a way to bias the MOSFETs that are used as a current source load. We can implement a current mirror since it would cause the current source load to pull the same amount of current as the mirror, which would allow us to restrict the current below the maximum allowed value for each branch. This would also allow us to mirror the current to multiple transistors, allowing for a simpler biasing implementation.

The implemented current mirror uses a PMOS with a resistor. The resistor determines the reference current, in other words, the current to be mirrored. The PMOS would then allow the current to be mirrored to other PMOS transistors, acting as a current source load.

Stage 1 and 2: Common Source Amplifier With Current Source Load

These two stages are designed to achieve a gain of at least 30dB; thus, when combined, they will achieve at least 60dB as per specifications. While the common source amplifier can achieve a noticeable gain, it can be further increased with a current source load implementation. The output of the first stage would go into the input of the second stage, separated by a coupling capacitor. This capacitor blocks the DC voltage, with AC being able to pass through, and since the voltage is not shared between the nodes, it allows biasing of the MOSFET to ensure it is in the saturation region. The first and second stages will be designed identically.

The current source load uses a PMOS biased with the current mirror. We choose to use a PMOS since the source/bulk would be connected to the highest potential (V_{DD}), and the drain of the PMOS is going towards the output. The current source load would further increase the gain compared to a regular resistor due to transconductance (G_m) being multiplied by a higher resistance value ($R_{O,1} || R_{O,2}$) compared to only being multiplied by the drain resistor R_D .

The rest of the common source uses an NMOS connected to the AC input at the gate, separated with a coupling capacitor. This means DC biasing is allowed since the DC will never reach the AC source, and thus, we can bias the gate of the NMOS with a voltage divider. We can not use the existing current mirror since the NMOS would require a separate voltage value at the gate to remain in the saturation region.

Stage 3: Source Follower

The goal of the buffer is to prevent massive load reductions to the overall gain. To meet the reduction of 10% or less (as per specifications), we must implement a buffer stage to mitigate the losses. The source follower (buffer) contains an NMOS, biased with a voltage divider and a source resistor and the bulk/body is connected to the lowest potential: ground. When a load is connected, it creates a voltage divider with the output; thus, if we use a buffer stage, the load looking into the second stage will become a very large value, which makes the voltage divider approximately equal to one. This would mean the gain would remain nearly identical due to the high resistance of the buffer stage.

It is also important to choose high resistances for the voltage dividers in stages 1-3, such that the current does not leak into the ground or source due to current favouring the path of least resistance.

Calculations

Current Mirror

Since we want to mirror the current to stages one and two, and the specifications require less than 1mW, we would need to keep the total current below $\sim 303\mu\text{A}$ ($303\mu\text{A} \cdot 3.3\text{v} = 0.9999\text{mW}$). Thus, we need to choose a reference current with respect to that.

Thus, let $I_D = 48.3 \mu\text{m}$.

Since the current mirror has a diode-connected PMOS, it would always be in saturation due to the threshold voltage being greater than zero. Now, let's find the range of possible gate voltages for PMOS saturation (receiving PMOS).

$$\begin{aligned} V_{SD} &\geq V_{OV} \\ V_S - V_D &\geq V_S - V_G - V_{tp} \\ 3.3\text{v} - V_D &\geq 3.3\text{v} - V_G - |-0.73\text{v}| \\ -V_D &\geq -V_G - 0.73\text{v} \\ V_G + 0.73\text{v} &\geq V_D \end{aligned}$$

Thus, a large gate voltage will ensure that the PMOS is running in saturation. Therefore, we can let $V_G = 1.98 \text{ V}$ to set V_{OV} within: $0.6 \geq V_{OV} \geq 0.2$ as per specifications.

To find the resistance, we can simply use Ohm's law.

$$R_{ref} = V_D / I_D$$

$$R_{ref} = 1.98\text{v} / 48.3\mu\text{A} = 41\text{k}\Omega$$

To find the width of the PMOS, we can let the length (L) = 5μm and use the current drain equation.

$$I_D = \frac{1}{2} k_p (W/L)(V_{OV})^2$$

$$48.3\mu\text{A} = \frac{1}{2} (25\mu\text{ A/V}^2)(W/5\mu\text{m})(3.3\text{v} - 1.98\text{v} - 0.73\text{v})^2$$

$$W = 55.5\mu\text{m (Width)}$$

Stages 1 and 2 (Identical)

The first amplifying stage contains the current to be mirrored, which depends on the aspect ratio of the current source load PMOS. For simplicity, we can let the ratio be 1:1, meaning the PMOS would have the same aspect ratio as the current mirror's PMOS: W = 55.5μm and L = 5μm.

To find the gate voltage to bias the NMOS, we first must find the range of acceptable gate voltages for saturation.

From PMOS,

$$V_{SD} \geq V_{OV}$$

$$V_S - V_D \geq V_S - V_G - V_{tp}$$

$$3.3\text{v} - V_D \geq 3.3\text{v} - V_G - |-0.73\text{v}|$$

$$-V_D \geq -V_G - 0.73\text{v}$$

$$1.98\text{v} + 0.73\text{v} \geq V_D$$

$$2.71\text{v} \geq V_D$$

Using this for NMOS,

$$V_{DS} \geq V_{OV}$$

$$V_D - V_S \geq V_G - V_S - V_{tn}$$

$$V_D - 0 \geq V_G - 0 - 0.63\text{v}$$

$$V_D + 0.63\text{v} \geq V_G$$

If we let V_D be half of the DC rail, $V_D = V_{DD}/2 = \sim 1.6\text{ V}$,

$$2.23\text{v} \geq V_G$$

From this, we can choose a value between $2.23\text{v} \geq V_G \geq 0$; thus, let $V_G = 0.854\text{ V}$ to set V_{OV} within: $0.6 \geq V_{OV} \geq 0.2$ as per specifications.

Now to bias the gate using a voltage divider, knowing $V_G = 0.854\text{ V}$, and if we let the resistor connected to V_{DD} be a high value: 3 MΩ.

$$[R/(3\text{M} + R)](3.3\text{v}) = 0.854$$

$$R = 1047424.37\Omega$$

Now, to find the width, we can let $L = 3\mu\text{m}$ and use the current drain equation.

$$I_D = \frac{1}{2} k_p (W/L)(V_{GS} - V_{tn})^2$$

$$48.3\mu\text{A} = \frac{1}{2} (100\mu\text{ A/V}^2)(W/3\mu\text{m})(0.854 - 0 - 0.63)^2$$

$$W = 54.3 \mu\text{m}$$

For small signal analysis, using the pi model, we found,

$$r_{O2} = r_{O1} = 1/(\lambda I_D) = 1/((0.04)(48.3\mu)) = 517598.3 \Omega$$

$$G_m = 2I_D/V_{OV} = 2(48.3\mu)/0.224 = 0.00043125 \text{ S}$$

And the gain $A_V = V_o/V_i$ to be,

$$A_V = -g_m (r_{O1} \parallel r_{O2}) = -0.00043125(258799.2) = -111.607 \text{ V/V}$$

$$A_{V, \text{dB}} = 20\log_{10}(A_V) = 20\log_{10}(|-111.607|) = 41 \text{ dB}$$

Thus, we are slightly past 40dB as per specifications due to the usage of current source biasing, which causes the gain per stage to increase significantly.

Since stages 1 and 2 are for amplification and they are identical, the total gain would be as follows,

$$A_{V, \text{total}} = (41\text{dB})(2) = 82 \text{ dB}$$

This would mean the total gain is past 60dB as per specifications.

Stage 3

To bias the NMOS, we must choose a value that puts it in saturation.

$$V_{DS} \geq V_{OV}$$

$$V_D - V_S \geq V_G - V_S - V_{tn}$$

$$3.3\text{v} - V_S \geq V_G - V_S - 0.63\text{v}$$

$$\text{Let } V_S = V_{DD}/2 = \sim 1.6\text{v}$$

$$3.3\text{v} - 1.6\text{v} \geq V_G - 1.6\text{v} - 0.63\text{v}$$

$$3.3\text{v} \geq V_G - 0.63\text{v}$$

$$3.93\text{v} \geq V_G$$

Thus, since voltage cannot go past 3.3 volts (due to the DC source), V_G can be any value within the DC rails, which means we can let $V_G = 2.53\text{v}$ to set V_{OV} within: $0.6 \geq V_{OV} \geq 0.2$ as per specifications.

To find the resistors for voltage divider biasing, we can let the resistor connected to V_{DD} equal $1\text{M}\Omega$ and find the bottom resistor.

$$3.3v(R/(R+1M\Omega)) = 2.53v$$

$$R = 3285714.3\Omega$$

Since the total current can not pass $303\mu A$, and we have used $3(48.3\mu A)$, we would have $158.1\mu A$ left to use before we fail the specification requirements. This means we can set the source resistor to pull a low current, where the leftover current goes towards the load.

Let $I_{Source} = 32.7\mu A$.

$$R_{Source} = 32.7\mu A (1.6v) = 52.32k\Omega$$

To allow for a more demanding load (smaller load resistance), we can set the drain current equation to have a potential load current as $I_D = 60\mu A$.

$$I_D = \frac{1}{2} k_p (W/L)(V_{GS} - V_{tn})^2$$

$$60\mu A = \frac{1}{2} (100\mu A/V^2)(W/3\mu m)(0.854 - 1.6 - 0.63)^2$$

$$W = 40 \mu m$$

Bandwidth

To find bandwidth, we can use the formula: $1/[2\pi R_{Node} C_{Node}]$. Thus, if we take the node between stage 2 and the buffer, we get,

$$R_{o1, stage 2} = R_{o2, stage 2} = 517.6k\Omega, R_{o, stage 2} = (517.6 k\Omega \parallel 517.6 k\Omega) = 258.8k\Omega; R_{Bias} = (3 M\Omega \parallel 1047424 \Omega) = 766.7 k\Omega; R_{o, buffer} = 1/\lambda I_D = 1/(0.04 1/V)(60 \mu A) = 764525\Omega; R_{Node} = \text{parallel of previous listed} = 154417 \Omega$$

$$C_{Node} = C_{GD, NMOS} + C_{GD, PMOS} + C_{GD, Buffer} + C_{GS, Buffer} = C_{ov}(W) + C_{ov}(W) + C_{ov}(W) + 2/3(C_{ox})(W)(L) = (0.4 fF/\mu m)(55.5 \mu m) + (0.4 fF/\mu m)(54.3 \mu m) + (0.4 fF/\mu m)(55.5 \mu m) + (2/3)(4.3 fF/\mu^2 m^2)(5 \mu m)(55.5 \mu m) = (22.2 + 21.72 + 22.2 + 1988.75) fF = 2054.87 fF$$

$$f_{-3dB} = 1 / [(2\pi)(154417 \Omega)(2054.87 fF)] = 501579.717 Hz = 501.580 KHz$$

Capacitance

When we combine the stages, they are separated by coupling capacitors that block DC and allow AC to flow through. To choose the coupling capacitance, we can set it to a high enough value so that the cutoff frequency approaches zero, allowing low-frequency AC signals to pass. Thus, we can set the coupling capacitance to $2\mu F$.

Simulation Data

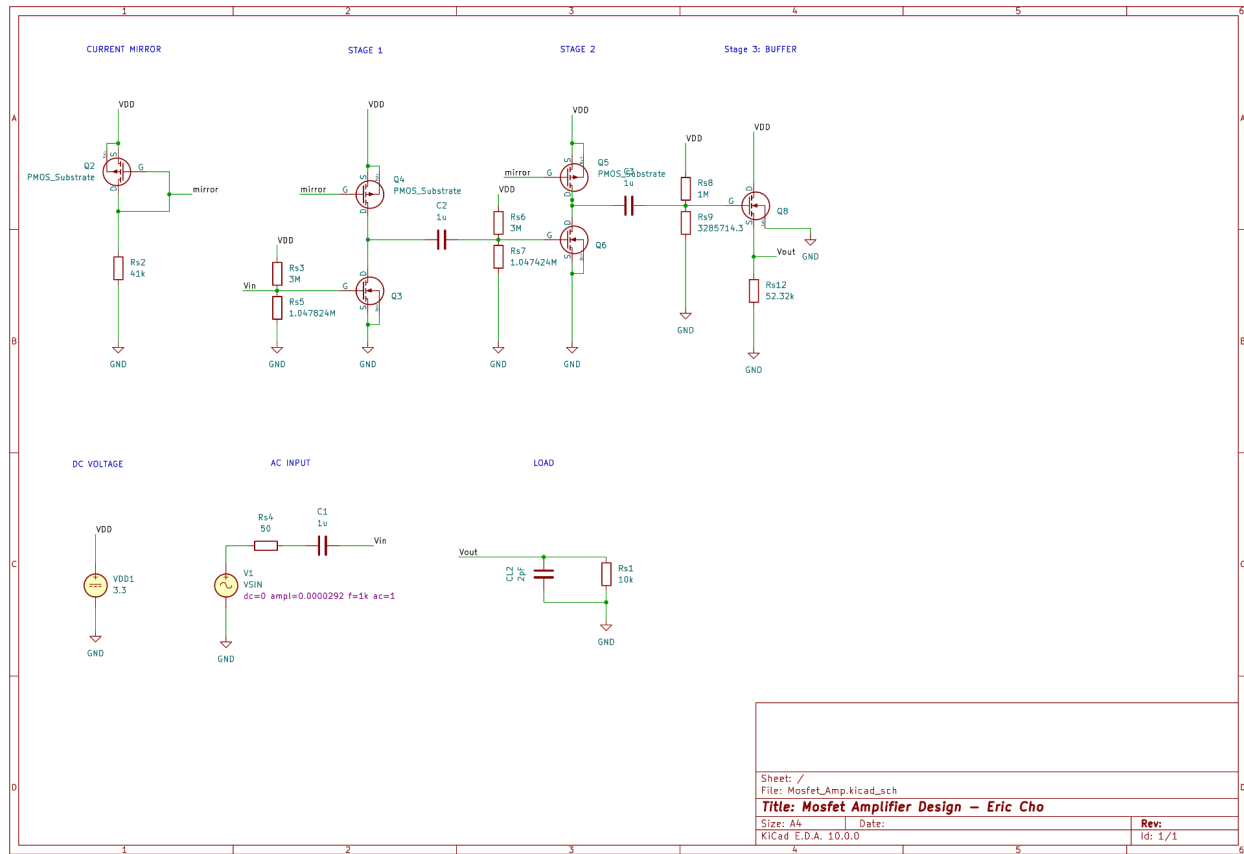


Figure 1: MOSFET Amplifier Connected With Labels

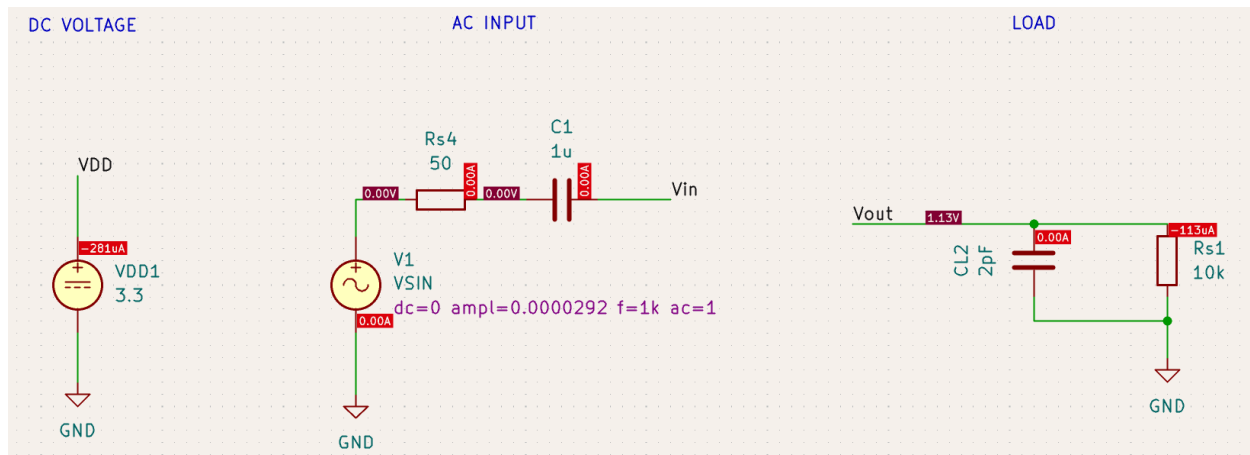


Figure 2: Current With Load

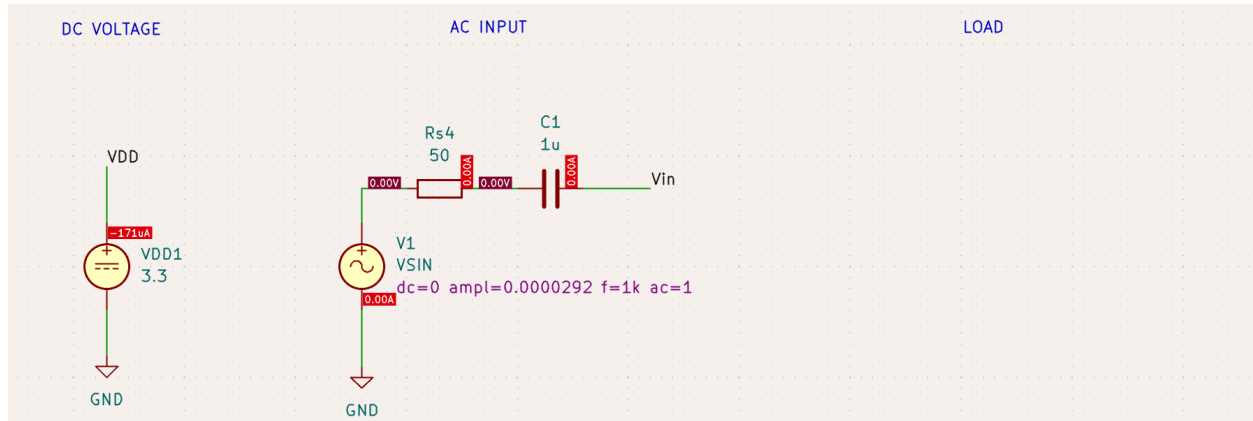


Figure 3: Current Without Load

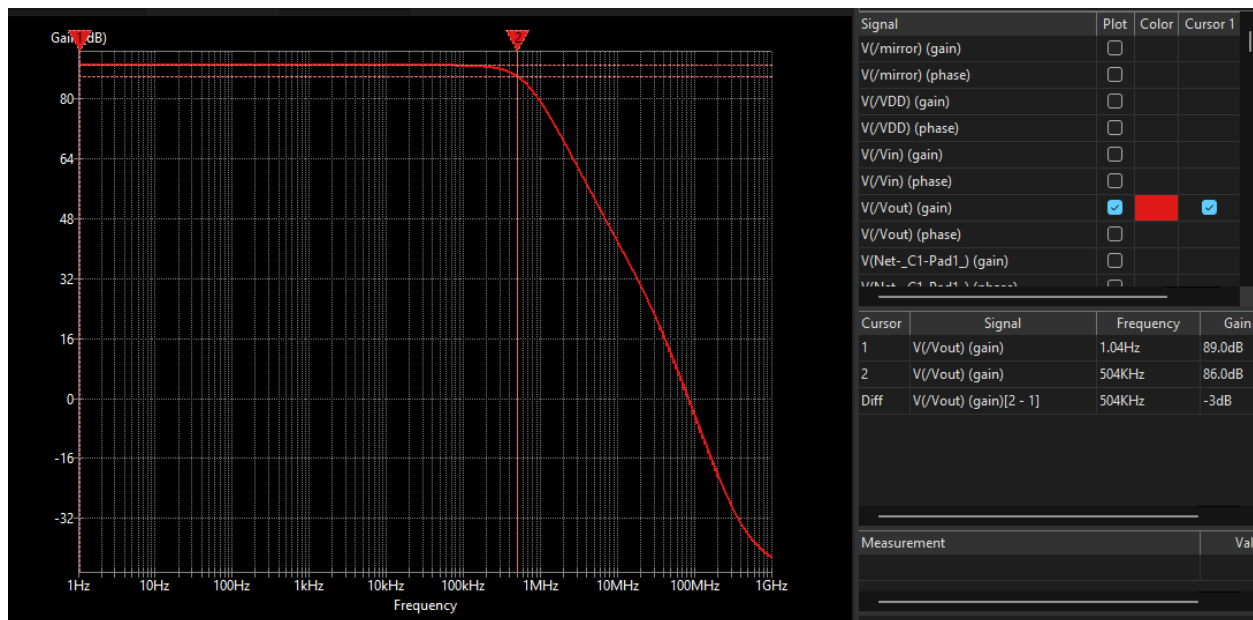


Figure 4: Gain and Bandwidth Shown With Cursors With Load

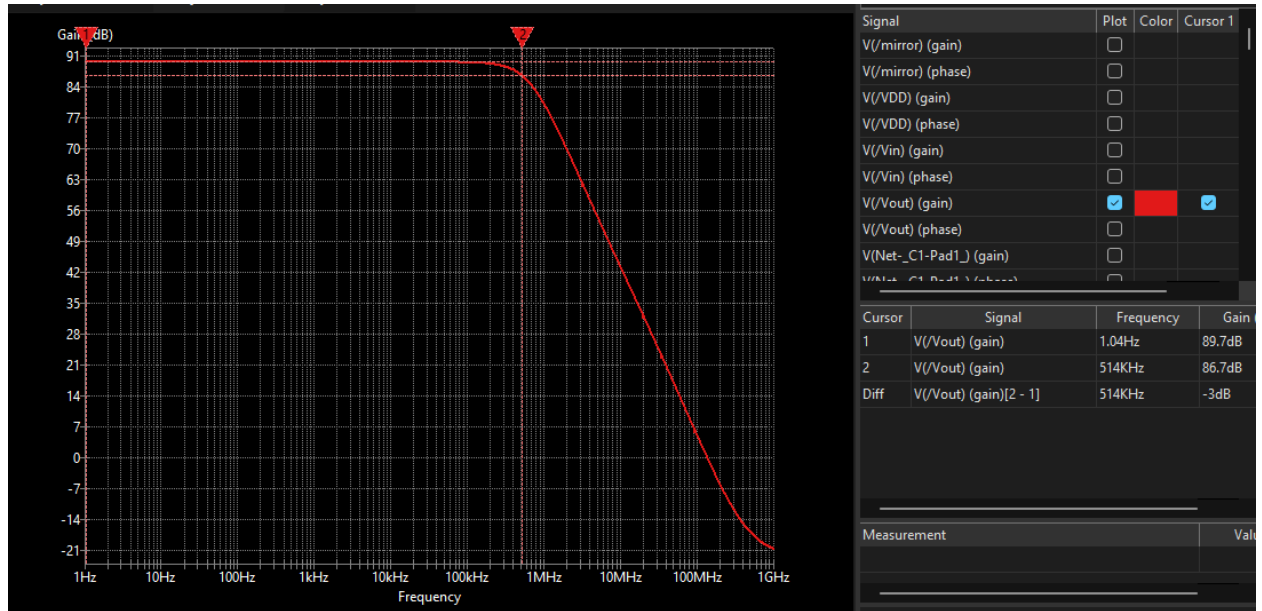


Figure 5: Gain and Bandwidth Shown With Cursors Without Load

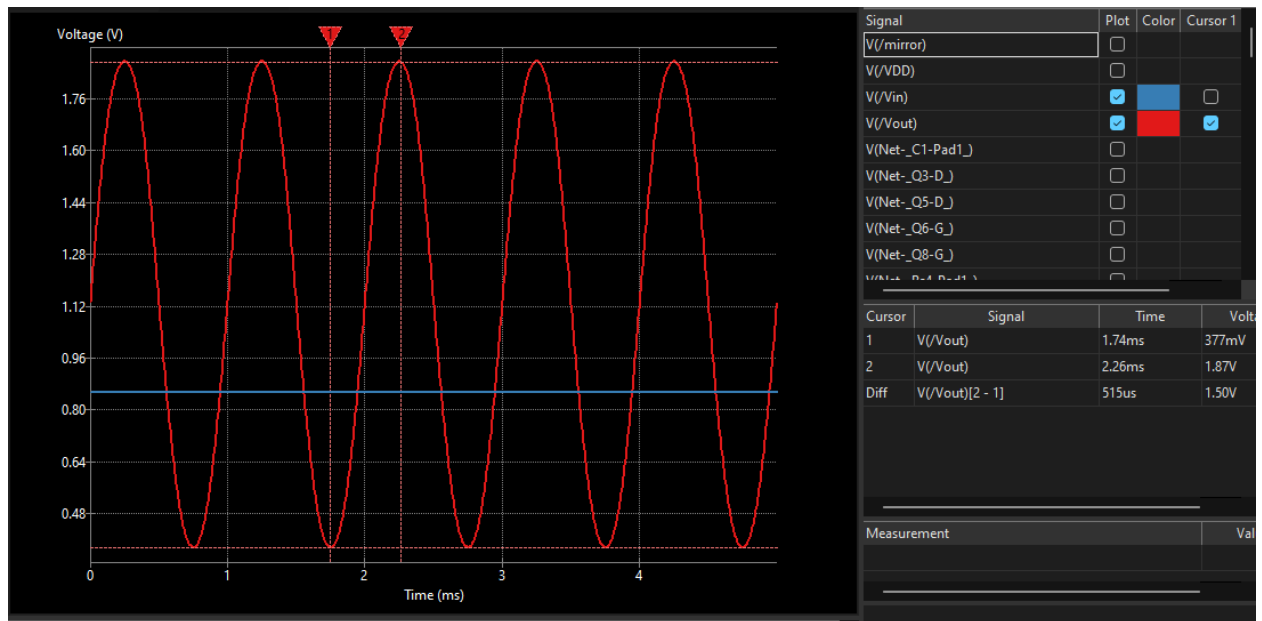


Figure 6: Waveform of the MOSFET Amplifier Output (red) with Input (blue)

Discrepancies and Discussion

Achieving Specifications in Simulations	
Gain	From Figure 4, the gain is above 60dB, reaching 89dB
Gain Reduction	From Figures 4 and 5, the %difference is $(1 - (89\text{dB}/89.7\text{dB})) \times 100 = 0.78\%$
Gain Per Stage	Per stage = total gain/2 = $89/2 = 44.5 \text{ dB}$ (since they are identical)
Total DC Power	From Figure 2: total current: $\text{Power}_{\text{DC, total}} = 3.3\text{V}(281\mu\text{A}) = 0.9273\text{mW}$
Max Branch Current	Current _{Mirror, Stage 1, Stage 2, Stage 3} = $47.8\mu\text{A} + 47.8\mu\text{A} + 47.8\mu\text{A} + 135\mu\text{A}$ (with load) Current _{Mirror, Stage 1, Stage 2, Stage 3} = $47.8\mu\text{A} + 47.8\mu\text{A} + 47.8\mu\text{A} + 25.2\mu\text{A}$ (no load)
Bandwidth	From Figure 4: $f_{-3\text{dB}} = 504 \text{ KHz}$ (with load) From Figure 5: $f_{-3\text{dB}} = 514 \text{ KHz}$ (without load)
Output Swing	From Figure 6, $V_{\text{Peak-to-Peak}} = 1.50\text{V}$.

Figure 7: Data Relevant to Amplifier Specifications

From Figure 7, we can see that most of the data achieved in the simulations matched the calculated results, except for the current drawn by the current mirror/buffer, the gain per stage, and the width of the NMOS transistors in stages 1 and 2. Since the current drawn by the current mirror differed by a slight amount (47.8μ vs. 48.3μ), the aspect ratio of the NMOS transistors needed to be adjusted to achieve the 1.6V target at the drain. The adjustment of the NMOS, specifically the width, is the only part changed from the manual calculations. This adjustment could have also affected the gain per stage, where we saw a 3.5 dB gain per stage increase over the calculated 41dB. When under load, the current drawn by the buffer increases from $25.2\mu\text{A}$ to $135\mu\text{A}$. To maintain a certain voltage at the output, the buffer pulls more current depending on the load connected. If we adjust the load, we can see that the lower the resistance, the higher the current, with higher resistance needing lower current. However, even with the increased power draw, the total current remained under $303\mu\text{A}$ at $281\mu\text{A}$ drawn. From Figure 5, we can see that the gain for a frequency between 0Hz and 504KHz (at -3 dB) will receive a gain of $\sim 89\text{--}87\text{dB}$. As seen in Figure 6, the input source having an amplitude of 0.0000292 and 1000Hz is being amplified to a signal that is reaching $1.5V_{\text{pp}}$. This means that the input is being increased by $\sim 89\text{dB}$, or in other words, a 28183.8 times increase.

Conclusion

As a final result, we were able to meet the specifications and create an amplifier that amplifies a small signal into a much larger signal that can be used for various applications.